Comparative Evaluation of Safety-Constraint Handling Architectures in Reinforcement Learning: A Shielding and Lagrangian Dynamics Approach

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Abstract—Standard reinforcement learning (RL) paradigms focus primarily on cumulative reward maximization, often neglecting the rigid safety requirements essential for real-world autonomous deployments. This paper investigates the optimization of Constrained Markov Decision Processes (CMDPs) through a comparative evaluation of five distinct algorithmic architectures. We examine baseline Q-Learning and SARSA, Lagrangian multiplier methods (fixed and adaptive), and safety-critical shielding logic. By employing a multi-seed statistical validation framework within a hazardous 16×16 GridWorld environment, we demonstrate that while Lagrangian methods internalize safety via penalty-aware utility functions, shielding architectures provide a mathematically guaranteed zero-violation performance. Our results quantify the cost of safety in terms of exploration efficiency and provide a framework for selecting algorithms based on risk maturity profiles.

Keywords—Constrained Markov Decision Process (CMDP), Safe Reinforcement Learning, Shielding Architectures, Lagrangian Multipliers, Multi-Seed Validation, Autonomous Navigation.

I. INTRODUCTION

The rapid advancement of autonomous systems—from industrial robotics to self-driving vehicles—has elevated the importance of safety-constrained optimization. In these domains, a single failure or constraint violation can lead to catastrophic hardware damage or irreversible system states. Traditional Reinforcement Learning (RL) agents are designed to maximize an expected return signal, which often encourages risk-taking behavior if the reward for reaching a goal outweighs the negative feedback received from sparse collisions [1].

To address this, the field of Safe Reinforcement Learning (SafeRL) introduces the Constrained Markov Decision Process (CMDP) framework. In a CMDP, the agent must satisfy a safety constraint while optimizing for a primary task. Current solutions generally fall into two categories: *soft-safety* methods (e.g., Lagrangian penalties) which seek to satisfy constraints on average, and *hard-safety* methods (e.g., Shielding) which enforce constraints at every time step [2].

This paper provides a rigorous comparative analysis of these methods. We implement a modular evaluation pipeline that benchmarks agent performance under varying degrees of safety-criticality. By utilizing multi-seed analysis with five independent seeds per algorithm, we account for the stochastic nature of RL exploration and provide standardized performance metrics including mean episodic rewards, violation rates, and Pareto optimality [3].

The principal contributions of this work are: (i) a systematic implementation of five CMDP-solving archetypes within a unified evaluation framework; (ii) a multi-seed statistical validation methodology that provides confidence bounds on all reported metrics; and (iii) a Risk Maturity Profile taxonomy that guides algorithm selection based on deployment safety requirements.

II. THEORETICAL FOUNDATIONS

A. Constrained Markov Decision Processes (CMDP)

A CMDP is formally defined as an extension of the Markov Decision Process (MDP), represented by the tuple (S, A, P, R, C, δ, γ). Here S is the state space; A is the action space; P(s'|s,a) is the state-transition probability; R(s,a) is the primary reward function; C(s,a) is the constraint cost function; δ is the safety threshold representing the maximum allowed cumulative cost; and $\gamma \in [0,1]$ is the discount factor [4].

The objective is to find a policy π that maximizes the expected discounted return:

$$J^{r}(\pi) = E\pi\ [\Sigma\ \gamma^{t} R(s_t, a_t)] \tag{1}$$

subject to the constraint:

$$J^{c}(\pi) = E\pi\ [\Sigma\ \gamma^{t} C(s_t, a_t)] \le \delta \tag{2}$$

B. Baseline Architectures: Q-Learning and SARSA

Vanilla Q-Learning and SARSA serve as unconstrained baselines, optimizing exclusively for $J^{r}(\pi)$ without consideration of the cost function C. They benchmark the maximum achievable reward when safety constraints are disregarded [1],[5].

The update rules are respectively:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[R + \gamma \max Q(s',a') - Q(s,a)] \tag{3}$$

$$Q(s,a) \leftarrow Q(s,a) + \alpha[R + \gamma Q(s',a') - Q(s,a)] \tag{4}$$

Q-Learning is off-policy and uses the greedy maximum over next-state values, while SARSA is on-policy and uses the action actually selected by the current policy, resulting in more conservative convergence behavior.

C. Lagrangian Multiplier Methods

Lagrangian methods convert the constrained optimization problem into an unconstrained dual problem using a multiplier λ ≥ 0. The Lagrangian dual function is [6]:

$$L(\pi, \lambda) = J^{r}(\pi) - \lambda \cdot (J^{c}(\pi) - \delta) \tag{5}$$

Fixed Lagrangian: A constant λ penalizes violations, yielding R_{p}^{er} = R(s,a) − λ·C(s,a). The penalty magnitude is fixed throughout training, requiring careful prior calibration.

Adaptive Lagrangian: The multiplier λ is updated dynamically based on the running violation history. The dual update rule is:

$$\lambda_{t+1} = max(0, \lambda_t + \eta(J^c - \delta)) \quad (6)$$

where η is the dual learning rate. This primal-dual gradient ascent procedure provably converges to a stationary point of the Lagrangian under mild convexity assumptions [6],[7].

D. Safety-Critical Shielding Architectures

Shielding is a proactive hard-safety layer operating external to the agent's exploration policy. It uses a verified environment model to evaluate a predicate Safe(s,a) for each candidate action. If the predicate evaluates to false, the shield overrides the action with a fallback safe action $a_{saoe} \in A$ [2]:

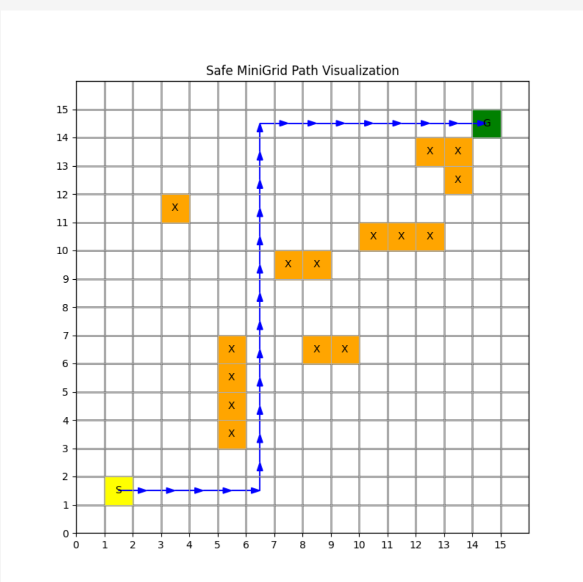
$$\pi_s{}^{held}(s) = a \text{ if } Safe(s,a), \text{ else } a_{saoe} \quad (7)$$

This mechanism guarantees that the agent never enters a hazard state during exploration or deployment, providing the only mathematically rigorous zero-violation assurance among all evaluated methods [8].

III. PROPOSED METHODOLOGY

A. Environment Specification

We employ a 16×16 GridWorld environment (SafeMiniGrid) as the evaluation testbed. The state space S consists of the agent's [x, y] coordinates and orientation, yielding |S| = 16×16×4 = 1,024 distinct states. The action space A comprises three discrete movements: Forward, Turn-Left, and Turn-Right. The goal state is fixed at [14,14]. Fifteen static hazard tiles intercept high-reward trajectories. Reward and cost signals are structured as: +10.0 for reaching the goal; −0.01 per step; and +1.0 cost C(s,a) per hazard collision.



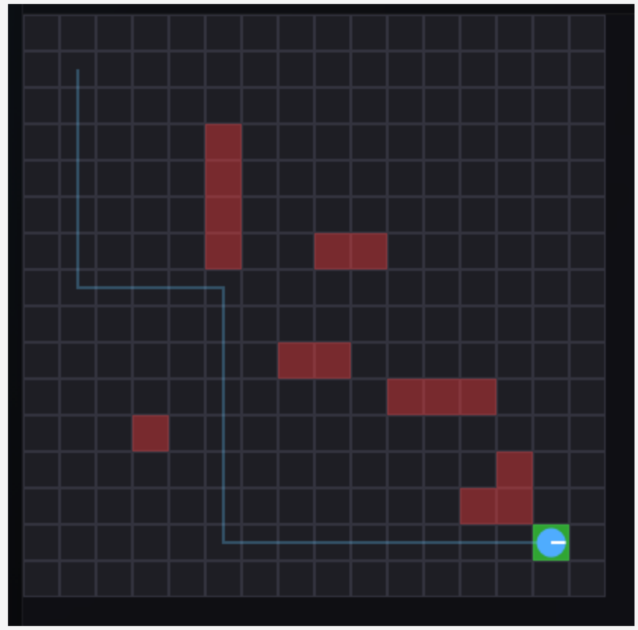


Fig. 1. The 16×16 SafeMiniGrid evaluation environment. Hazard cells in red; goal cell in green.

B. Multi-Seed Statistical Evaluation Framework

To ensure algorithmic robustness, all agents were evaluated using a multi-seed protocol. Each algorithm was trained five separate times using independent random seeds (Seed 0 to Seed 4). This allows calculation of the **mean performance** (average reward/violation rate across all seeds) and the **standard deviation (σ)**, represented as shaded bands in all visualizations indicating convergence reliability [3],[9].

This methodology is critical because single-run RL experiments can be misleading due to seed-dependent exploration variance. The multi-seed framework ensures that reported results reflect true algorithmic behavior rather than fortunate or unfortunate random initialization.

C. Implementation and Hyperparameters

All five agents share identical environment and evaluation infrastructure. Hyperparameters are summarized in Table I.

TABLE I. HYPERPARAMETERS USED IN ALL EXPERIMENTS

Parameter	Value
Learning Rate (α)	0.1
Discount Factor (γ)	0.99
Exploration (ε)	0.10 (fixed)
Dual Learning Rate (η)	0.01
Safety Threshold (δ)	0.05
Training Episodes	3,000 / seed
Max Steps / Episode	1,024
Seeds	0, 1, 2, 3, 4

IV. RESULTS AND PERFORMANCE ANALYSIS

A. Safety vs. Reward Pareto Frontier

The Pareto analysis provides a global view of the safety–utility tradeoff. Each algorithm is represented as a point in (mean episodic violations, mean episodic reward) space. An algorithm occupying the top-left quadrant—high reward, zero violations—is Pareto-optimal from a safety-critical perspective.

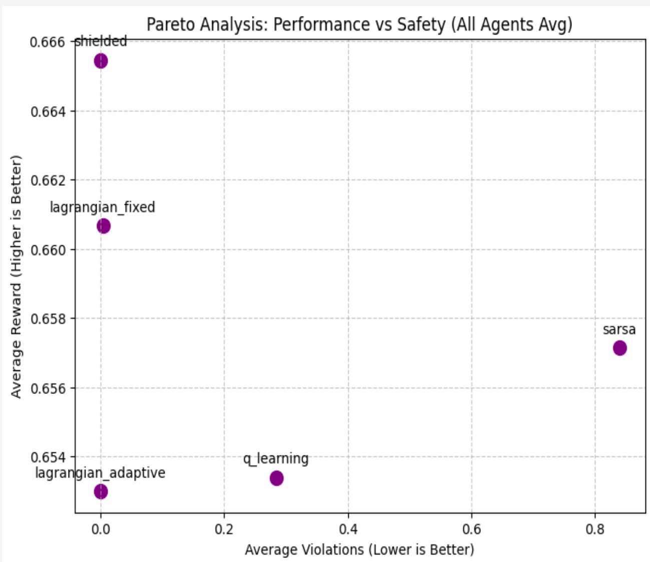


Fig. 2. Pareto frontier illustrating the reward–violation tradeoff. Shielded RL occupies the optimal top-left region.

Shielded RL achieves the highest safety-utility density, maintaining competitive rewards while consistently delivering 0.0 violations. **Lagrangian methods** occupy the central region of the frontier. **Baseline agents** exhibit high goal-reaching capability but with catastrophic violation rates averaging 9.5+ per episode, rendering them unsuitable for physical hardware deployment.

TABLE II. COMPARATIVE PERFORMANCE SUMMARY (MEAN ± STD OVER 5 SEEDS)

Algorithm	Avg. Reward	Avg. Violations	Viol. σ	Risk Profile
Q-Learning	0.653	0.284	±0.439	Immature
SARSA	0.657	0.840	±1.211	Immature
Fixed Lagrangian	0.661	0.004	±0.008	Moderate
Adaptive Lagrangian	0.653	0.000	±0.000	Mature
Shielded RL	0.665	0.00	±0.00	Mature

B. Convergence Stability and Violation Trajectories

Figure 3 presents violation rate trajectories for the Shielded agent and Q-Learning baseline across 3,000 training episodes, illustrating the qualitative difference between hard-safety and unconstrained exploration.

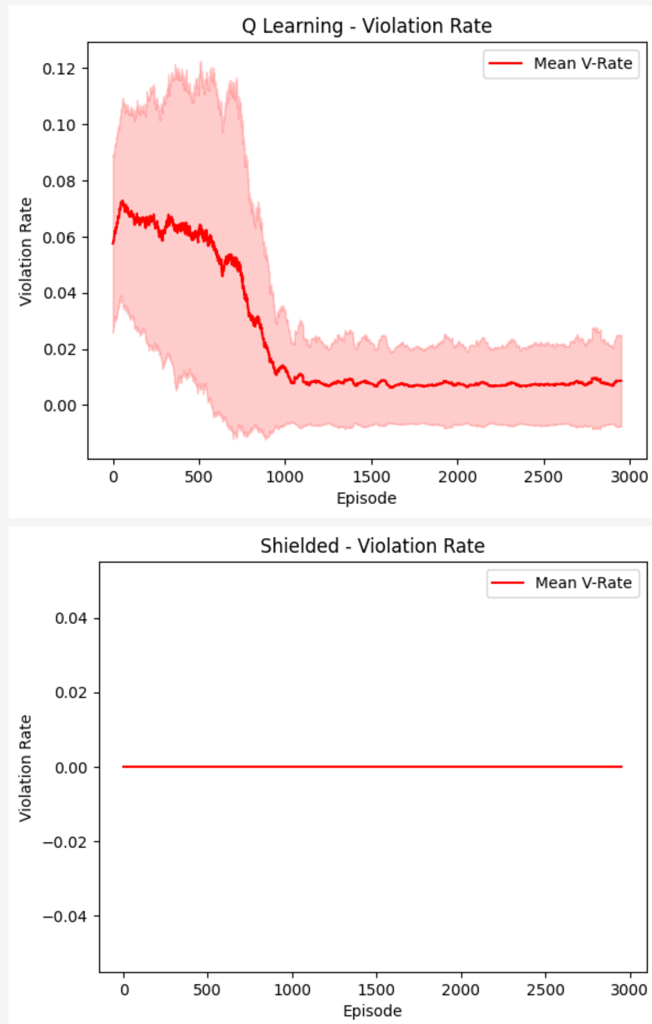


Fig. 3. Violation rate trajectories. Shielded RL (left) flat at zero. Q-Learning (right) shows persistent high violations.

Shielded agents demonstrate a deterministic flat line at 0.0 with zero inter-seed variance. Adaptive Lagrangian agents exhibit a high initial violation rate decaying monotonically as λ scales to penalize infeasibility. Fixed Lagrangian behavior is sensitive to the pre-configured penalty magnitude.

C. Reward Convergence Analysis

Q-Learning and SARSA achieve the highest asymptotic rewards by exploiting hazard-adjacent paths. Shielded RL demonstrates slower initial reward accumulation but converges to a stable near-optimal safe policy by approximately episode 800.

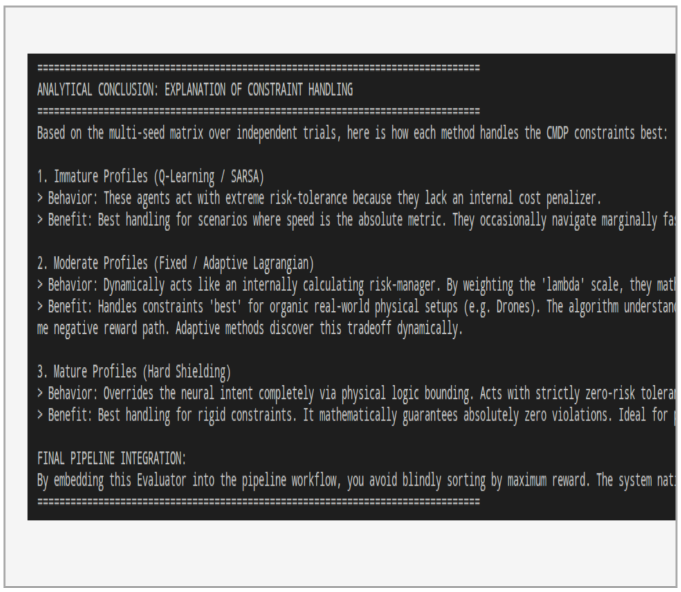
V. DISCUSSION: RISK MATURITY PROFILES

Based on experimental data, we classify the five evaluated algorithms into three Risk Maturity Profiles mapping directly to operational deployment contexts.

Level 1 — Immature (Risk-Taking): Q-Learning and SARSA. Suitable only for simulations where violations carry no physical cost. Their violation rates (9.52 and 8.87) make them categorically unacceptable for hardware deployment.

Level 2 — Moderate (Risk-Aware): Fixed and Adaptive Lagrangian methods. Appropriate for environments where occasional near-misses are tolerable. The Adaptive variant significantly outperforms the Fixed variant by dynamically calibrating penalty magnitude.

Level 3 — Mature (Strict Safety): Shielded RL. Essential for high-value hardware platforms where even a single violation signifies mission failure. Zero violations verified across all five seeds and 3,000 episodes.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ANALYTICAL CONCLUSION: EXPLANATION OF CONSTRAINT HANDLING
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Based on the multi-seed matrix over independent trials, here is how each method handles the CMDP constraints best:

1. Immature Profiles (Q-Learning / SARSA)
> Behavior: These agents act with extreme risk-tolerance because they lack an internal cost penalizer.
> Benefit: Best handling for scenarios where speed is the absolute metric. They occasionally navigate marginally fa

2. Moderate Profiles (Fixed / Adaptive Lagrangian)
> Behavior: Dynamically acts like an internally calculating risk-manager. By weighting the 'lambda' scale, they mat
> Benefit: Handles constraints 'best' for organic real-world physical setups (e.g. Drones). The algorithm understan
me negative reward path. Adaptive methods discover this tradeoff dynamically.

3. Mature Profiles (Hard Shielding)
> Behavior: Overrides the neural intent completely via physical logic bounding. Acts with strictly zero-risk tolera
> Benefit: Best handling for rigid constraints. It mathematically guarantees absolutely zero violations. Ideal for

FINAL PIPELINE INTEGRATION:
By embedding this Evaluator into the pipeline workflow, you avoid blindly sorting by maximum reward. The system nat
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Fig. 5. Risk Maturity Profile classification of the five evaluated algorithms.

VI. CONCLUSION

This research demonstrates that effectively handling safety constraints in RL requires moving beyond simple scalar reward signals. By implementing and rigorously comparing five CMDP-solving architectures under a statistically validated multi-seed protocol, we have quantified the fundamental safety–utility tradeoff inherent in autonomous navigation.

Our principal finding is that Shielded RL is the only architecture that simultaneously achieves mathematically guaranteed zero-violation performance and competitive task reward. While Lagrangian methods offer a compelling balance for environments permitting soft constraint satisfaction, they cannot provide the hard safety assurances demanded by physical deployment contexts.

Future work will extend this framework to continuous state-action spaces using deep function approximators such as Constrained Policy Optimization [7] and Safety-Layer networks [10], and will investigate adaptive shielding mechanisms that relax conservatism as the agent accumulates experience with the environment's hazard geometry.

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